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Information opacity and corporate bond returns: The dynamics of split ratings

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ABSTRACT

The disagreements among credit rating agencies (split ratings) change over time, reflecting time varying nature in information opacity of firms. We analyse how the changes in information opacity affect the confidence in the reliability of ratings. We document that negative credit rating adjustments have a stronger impact on bond excess returns: (i) when the information opacity cuts down and (ii) with lower levels of information opacity. Investors seems to have more trust on the information disclosed by rating adjustment in those cases. The excess return response also depends on the credit rating level, being the dynamics of opacity relevant only for high yield (HY) bonds. The key results persist when alternative measures of rating agencies' disagreement are taken into consideration.

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1. Introduction

Credit Rating Agencies (CRAs) are financial information providers that deliver their opinions on creditworthiness of debt issuers, i.e., the probability of repaying their debts, into a credit rating. Their important role in financial markets has been a topic of research for years (see [Matthies, 2013](#) for a review of the credit rating-related research). Although ratings are opinions and not recommendations to buy, sell or hold, they have a central role in credit markets. Financial regulations often rely on them and issuers use ratings as a signal of low investment risk to reduce their capital costs in debt markets.

Financial markets incentivise debt issuers to ask for more than one rating to reduce their borrowing costs ([Baker and Mansi, 2002](#)). When several CRAs rate the same issue, they may agree on the opinion about its creditworthiness, providing equivalent ratings. Otherwise, CRAs may disagree, providing different ratings (split rating).¹ Split rating provides a useful sig-

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E-mail addresses: pilar.abad@urjc.es (P. Abad), rodrigo.ferreras@santalucia.es (R. Ferreras), mdrobes@ucm.es (M.-D. Robles).¹ Split ratings are common in the US corporate bond market. [Cantor, Packer and Cole \(1997\)](#) find 57.4% of rating discrepancy from 1983 to 1993 and [Livingston and Zhou \(2010\)](#) a 48% in the 1983–2006 period.

nal for asymmetric information and opacity. According to Morgan (2002), the disagreement on the rating assigned by different CRAs arises due to the problem of information opacity of issuers. He models formally the relation between split ratings and information opacity and finds that higher uncertainty over the default risk of an issuer increases the probability of disagreement among CRAs.²

Previous research on the effect of split ratings on corporate bonds focuses exclusively on the date of issuance (Morgan, 2002; Jewell and Livingston, 1999; Livingston and Zhou, 2010). However, the information environment usually changes over time and CRAs react to the new degree of opacity by adjusting the assigned ratings if necessary. These adjustments are commonly asynchronous and of different magnitude, generating dynamic changes in the split rating along the lifetime of the bond. Although the disagreement among CRAs play a substantial role in reflecting the changing information environment of firms, it is largely ignored in earlier research. So, the study of the effect of these dynamics over the bonds' lifetime, instead of placing the focus on the issuance date, provides us a clear agenda.

Our first hypothesis is that bond return response to rating changes depends on the dynamics of information opacity proxied by the split rating. Previous literature shows that: (1) greater CRAs' disagreement is associated with higher cost of debt at issuance³ and (2) the existence of a negative corporate bond market response to the information disclosed by rating deteriorations⁴. Although CRAs facilitate the functioning of capital markets by reducing information asymmetries through rating announcements, CRAs' disagreements could mitigate the role of ratings in reducing those asymmetries.⁵

On this paper we study how changes in the firm opacity level (i.e., changes on the split rating) alter the investor's responsiveness to rating adjustments in a sample of 16,625 announcements of credit rating deterioration for US corporate issues by Moody's, S&P and Fitch (the Big Three) between 2004 and 2014. We focus on the excess return response to the split rating dynamics. Our principal finding supports our expectation about the mitigating effect of opacity. We document that bond return's response to negative rating adjustments is stronger when the split rating cuts down. Lower split ratings are signals of higher transparency about the creditworthiness of the issue, leading investors to increase their trust regarding the information disclosed by the CRAs' rating announcements.

Our second concern is how the effect of the split rating dynamics (to negative rating adjustments) on bond return depends on the information opacity level. According to Badoer and Demiroglu (2019), the information content of credit ratings relates with global market information transparency, affecting investors reliance on ratings. We expect that higher levels of uncertainty about the credit risk of a firm will lower investors reliance on the information that the rating adjustment discloses. We find evidence of a stronger effect of rating deteriorations on bond returns when the level of information opacity is lower in line with our second hypothesis.

We deepen the analysis by studying whether the credit risk level of the related bond is relevant to explain the effect of the split rating dynamics (our third hypothesis). We are concerned on the effect of CRAs disagreements on riskier bonds, as they are more sensitive to the information environment and uncertainty.⁶ Our evidence points out that the effect of information opacity on the return response to downgrades is reinforced for High Yield (HY) bonds. The uncertainty regarding the credit risk of more opaque bonds induces a worse investor perception for low-rated issues than for bonds with higher ratings.

Finally, to assess the robustness of our results we repeat the analysis with two different measures of CRAs' disagreement that consider the whole distribution of ratings assigned to the same bond instead the two extremes. Our key results are robust if we use the alternative methodology to measure the divergence of opinion between CRAs.

The paper is structured as follows: in Section 2, we explain the data used. In Section 3, we present the research design and Section 4 shows the empirical results. Section 5 introduces two alternative measures of CRAs' disagreements. We present the main conclusions in Section 6.

² Split rating is used to measure opacity for banks (e.g., Bannier, Behr and Güttler, 2009; Balasubramnian and Cyree, 2011), firms (e.g., Kim, Kraft and Ryan, 2013; Hyttinen and Pajarinen, 2008), different corporate issues (Bowe and Larik, 2014), insurance companies (Pottier and Sommer, 2006), corporate bonds (Güntay and Hackbarth, 2010; Livingston, Wei and Zhou, 2010) and loans (Drucker and Puri, 2008).

³ Livingston and Zhou (2010) and Livingston et al. (2010) find that the opacity of issuer information leads to bonds with split ratings to have an average excess spread of 7 basis points over bonds with similar credit risk and no split rating. Likewise, Livingston and Zhou (2010) note that on economic downturns the excess spread due to market opacity is higher. Duffie and Lando (2001) model theoretically how investors will seek compensation to hold risky assets when the firm value is not perfectly observable. Bonsall and Miller (2017), Mansi, Maxwell and Miller (2011), Lu, Chen and Liao (2010) and Gao, Wang, Wang, Wu and Dong (2020) find that a lack of transparency is associated with higher cost of debt. Platikanova and Soonawalla (2019) find that more opaque financial reporting hinders the creditors' ability to evaluate credit risk.

⁴ There is an extensive literature on the bond investors' response to rating changes. Some examples are Steiner and Heinke (2001), Dichev and Piotroski (2001), May (2010), Cornaggia and Cornaggia (2013). The common result is that negative rating adjustments convey new information as they cause a negative bond market response.

⁵ He, Wang and Wei (2010) find that information asymmetry changes with rating adjustments. Livingston, Naranjo and Zhou (2008) find that bonds issued by more opaque firms, i.e., with higher levels of split ratings, are prone to suffer from subsequent rating changes.

⁶ Jorion and Zhang (2007) find that the CRAs' opinion has a stronger effect for below-investment grade issues. Kwan (1996) shows that HY bonds are particularly sensitive to firm-specific information; Han and Zhou (2013) indicate that bonds that are closer to default are more sensitive to reductions of information and Hao, Prevost and Wongchoti (2018) argue that HY bonds have greater exposure to information risk.

2. Rating event selection and split rating dynamics

We analyse bond return responses to negative rating events. Following Gande and Parsley (2005), we use a comprehensive definition of credit rating events by considering rating downgrades and negative changes in the watch-status of bonds announced by at least one of the Big Three CRAs (S&P, Moody's and Fitch Ratings).⁷ Notice that a rating event may be announced by just one single CRA or various CRAs simultaneously (on the same date). We consider the events in the last case as one single event and denote them as multi-rating events.

We collect bond credit rating information by the Big Three rating agencies from Mergent Corporate Bond Security Data. According to the Securities and Exchange Commission (SEC), these agencies covered approximately 95% of the US market at the end of 2017 (SEC, 2017). We gather information on downgrades and changes in the watch listing status with negative connotations of corporate plain vanilla fixed rate bonds⁸ issued by US firms and denominated in US dollars from October 1st, 2004, to December 31st, 2014. We exclude ratings not tabulated in the common scales and short-term ratings. Second, we collect the daily prices of these re-rated bonds from TRACE.⁹ Following Dick-Nielsen (2009) and Edward, Harris and Piwowar (2007), we eliminate TRACE errors and information that is not useful for bond price analysis.¹⁰ Finally, we set an event-driven liquidity filter by requiring at least 4 traded prices between consecutive rating events on the same bond. This filter also avoids event window contamination due to the clustering of events (see the next section for further details).

Table 1 depicts the distribution of rating events in the final sample. It contains 16,625 downgrades and changes in the watch status. Regarding the announcing CRA (Panel A), 43% correspond to Standard & Poor's, 47% to Moody's and 17% to Fitch. In order to capture any signal of change in credit risk, we map the ratings into a 58-point numerical scale that considers the rating level as well as the watch status (Sy, 2004). The largest value is assigned to the best rating level, AAA = 58, the second is AAA watch - = 57, AA + watch + = 56 and so on until the worst rating level, CC to SD-D = 1.¹¹ Watch - (+) denotes additions to the watch list for review for downgrading (upgrading). A change in watch status implies a change of 1 point in the numerical scale whereas a rating change of one notch (for example, from A- to BBB +) implies a change of 3 points.¹² The average Standard and Poor's rating is 32.9 (BBB), which is 2 points higher than the average Moody's rating (31, BBB- watch +). This result is in line with the studies of Baker and Mansi (2002) and Livingston et al. (2008), corroborating that Moody's provides more conservative ratings than S&P in the sample period. The Fitch's average rating (33.9, BBB, watch positive) is the highest among the CRAs, which is consistent with Jewell and Livingston (1999), who finds that the average Fitch rating is much more positive than the ratings of Moody's and S&P. When comparing the rating assigned before the announcement, we observe that Fitch is the best rating provider (assigning lower credit risk) in 52% of the events. On the opposite side is Moody's, which provides the best rating in only 33% of the cases, closely followed by S&P. When comparing the worst rating providers (assigning a higher credit risk), Moody's is the strictest rating provider in 46% of the cases, closely followed by S&P with 45% of events and Fitch, which is the worst rating provider for 25% of events.¹³

In terms of the degree of rating deterioration (see Panel B), 58% of rating events correspond to a deterioration of 3 points or less in the numerical rating scale (the equivalent of one complete rating notch or less), 87% of the events are between 1 and 6 points and 6% of the events show more than 9 points of deterioration (more than one complete rating letter). The average rating drop is 4 points and is quite similar across CRAs.

Panel C of Table 1 shows the distribution by the number of CRAs that monitor the rated bonds. At the rating event date, 13% of the issues are followed by just one CRA, 23% are followed by two CRAs and 64% by three CRAs. 83% of the events are announced by a single CRA, while the remaining 17% are multi-rating events.

Fig. 1 presents the time distribution of rating events. Years 2008 and 2009 (the global financial crisis period) present the highest number of rating events (53%), while 2005, 2013 and 2014 are the years with fewer rating announcements, accounting altogether for 3% of the events.

Table 2 depicts the distribution of rerated bonds by relevant characteristics. After the filtering process described above, the number of different bonds in the sample is 6,157 from 489 different issuers, which are affected by an average of 2.7 rating events per issue. The average time to maturity is 3.84 years. Most of the bonds are senior issues, which account for a 91.1% of the sample and 15.9% of bonds contain covenants to enhance the credit profile of the issues.

⁷ CRAs frequently put corporate bonds on credit watch (positive or negative) and decide a few months later to change or to confirm the explicit credit rating. In our setting, we consider as rating events any change in the watch status of bonds with negative implications.

⁸ This excludes bonds with optionality or redeemable, convertible bonds, asset backed securities, issues linked to other issues and perpetual bonds.

⁹ TRACE is the Financial Industry Regulatory Authority (FINRA) vehicle to comply with the Rule 6200 series that requires a reporting of over the counter fixed income transactions. It started in January 2002 and covers almost 100% of the corporate bond market transactions since 2004.

¹⁰ We filter out: (i) true duplicates (identified by the same message sequence number), (ii) same day reversals (occurring on the same day or in any other day identified by the R or the X), (iii) operations cancelled and corrections (cleaning the previous to correction operation), (iv) other errors (traded prices over 500\$ or below 0.001, trade volumes below or equal to 0 or where there is no transaction time), (v) transactions indicated as "when issued", (vi) agency transactions (double reports matching in most data, but not identified by the same message sequence number) identified by considering operations on the same instrument with a delay in the reporting time of up to 3 min and with a price difference of 5% considering more than one reporting party, and (vii) outliers (returns from the previous price in absolute terms greater than 20% and prices above 20% in absolute terms with respect to the median daily price). After this filtering, we compute the daily price as the average price of the day.

¹² Notice that there are simultaneous changes in the rating level and in the watch status. So, for instance, a change from A- watch negative to BBB+ implies a change of 2 points.

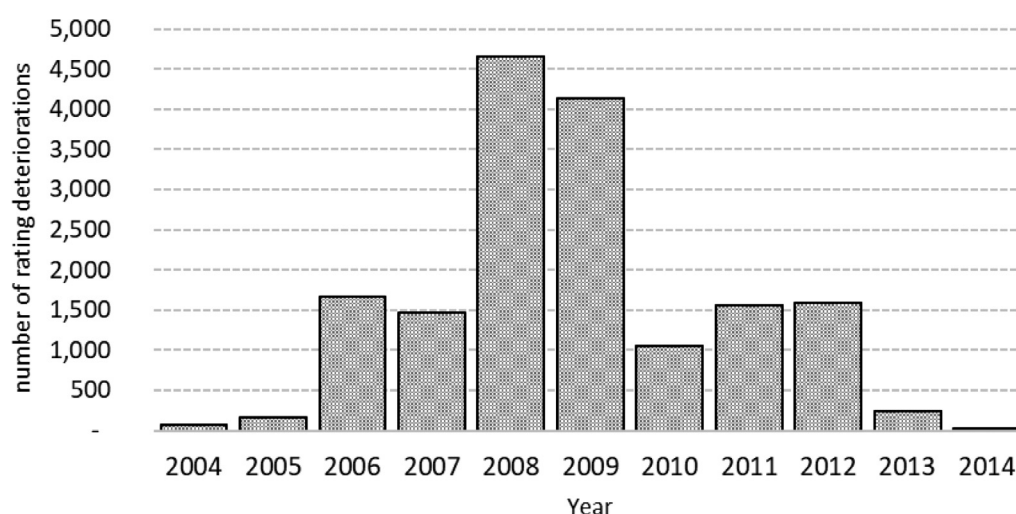
¹³ Notice that the rating assigned by the different CRAs can be the same at the event date.

Table 1

Distribution of rating events.

Panel a. Credit rating agency				
CRA	S&P	Moody's	Fitch	
% events by CRA	43.4%	47.0%	17.4%	
Average CRt	32.9	31.0	33.9	
Average ΔCR	4.4	4.4	4.1	
% Best rater	33.5%	30.6%	48.6%	
% Worst rater	42.1%	42.6%	23.2%	
Panel b. Rating change size				
Jump size	ΔCR ≤ 3	ΔCR ≤ 6	ΔCR ≤ 9	
% events	58%	87%	94%	
Panel c. Number of CRA following				
# CRA following the issue (issuer)	1	2	>2	
% followed by	13.0%	23.5%	63.6%	
% announced simultaneously by	82.9%	13.4%	3.7%	

A rating event is a downgrade or a change in the watch status of a bond. The total number is 16,625 rating events from October 1st, 2004, to December 31st, 2014, where 3,598 are changes in watch status (changes of 1 or 2 points on the 58-point numerical scale). CR is the credit rating of the bond on the event day, ΔCR is the size of the event defined as $\Delta CR = \text{abs}(CR - CR_{-1})$, where the subscript -1 indicates the day before the event. *Best (Worst)* rater indicates the percentage of events on which the CRA assigns the highest (lowest) rating on the event date. The total percentage of events by the CRA is larger than 100% due to multi-rating announcements. In some cases, several CRAs assign the same best (worst) rating, so the corresponding sum of percentages is above 100%.

**Fig. 1.** Distribution of negative rating events by year. Note: 16,625 rating deterioration announcements from October 1st, 2004, to December 31st, 2014.**Table 2**

Characteristics of rerated bonds and issuers.

# different issues	6,157
Number of issuers	489
Average years to maturity	3.84
# downgrades affecting to senior issues	15,085 (84.19%)
# downgrades affecting to issues with covenants	2,644 (14.76%)

The sample comprises 16,625 credit rating deteriorations announced from October 1st, 2004, to December 31st, 2014. # is the number, % is the percentage of the total number of events.

We measure the daily split rating for bonds rated by more than one CRA as the difference between the higher and the lower ratings assigned by the different CRAs to the same bond each day as follows:

$$\text{Split}_{b,t} = \text{Max}(CR_{S\&P,b,t}, CR_{\text{Moody's},b,t}, CR_{\text{Fitch},b,t}) - \text{Min}(CR_{S\&P,b,t}, CR_{\text{Moody's},b,t}, CR_{\text{Fitch},b,t}) \quad (1)$$

where $CR_{j,b,t}$ is the rating assigned to the bond b on day t by agency j ($j = \text{S\&P, Moody's, Fitch}$).

The split rating is dynamic, as it changes when a CRA announces an adjustment to the rating for this bond. Table 3 depicts the distribution of split ratings and provides the first view regarding the level of CRAs' disagreement, i.e., the differences in

Table 3

Split rating distribution.

	#	(%)	Mean	Median	Max	Min	Std
ΔCR	16,625	100%	4.31	3	40	1	3.02
Split	13,791	83%	6.63	6	44	1	4.44
Split-1	13,803	83%	5.77	5	44	1	4.15
ΔSplit	12,212	73%	0.97	2	40	-24	4.16
ΔSplit	12,212	73%	3.36	3	40	1	2.63
ΔSplit = 0	4,413	27%					
ΔSplit < 0 (Narrow)	4,757	29%	-3.07	-3	-1	-24	1.94
ΔSplit > 0 (Widen)	7,455	45%	3.55	3	40	1	2.98

The sample comprises 16,625 credit rating deteriorations announced from October 1st, 2004, to December 31st, 2014. CR_{i-1} is the prior rating level by the CRA announcing the event, where the subscript -1 indicates the day before the event. ΔCR is the size of the event, defined as $\Delta CR = \text{abs}(CR - CR_{i-1})$. $Split$ is the maximum discrepancy between the ratings assigned by the CRAs at the announcement date; $Split_{i-1}$ is the split level prior the announcement. $\Delta Split$ is the change in the split level, defined as $\Delta Split = Split - Split_{i-1}$. # Indicates the number of events, % indicates the percentage with respect to the total number of events.

their assessments of bond credit risk. The average split rating is 5.5 points (almost 2 complete notches), presenting a discrepancy level of 4.79 points before the event announcement. The average change in a split rating on the event date is 0.97 and the average of the absolute value of the split rating change is 3.36. A total of 74% of the 16,625 rating events lead to a change in the disagreement between CRAs' opinions. They lead to a more convergent opinion on the credit worthiness of the issue, that is, a narrowing of the split rating in 28.6% of the events and to a more divergent opinion, that is, a widening of the split rating in 44.8% of the events. These split rating dynamics imply a general increment of information opacity in the US corporate bond market in this period and give some indirect evidence against the herding behaviour hypothesis (see Lugo, Croze and Faff, 2015).

3. Research design

To analyse the effects of information opacity on the returns of rerated bonds, we first calculate the bond returns from clean prices. This allows us to avoid contamination of the results due to daily mechanical increases or sudden decreases in gross prices from accrued interest (see Defond and Zhang, 2014).¹⁴ We calculate the bond return around a rating event i from the last daily price before the event date (at day $t = t_1$) to the first daily price after it (at day $t = t_2$).¹⁵ This definition of the event window allows us to increase the sample size and the power of the tests (see Ederington, Guan and Yang, 2015). Following Bessembinder, Kahle, Maxwell and Xu (2009), we compute the premium event period return (excess return for short) around the event i , by subtracting the return of a matched Treasury security with a similar maturity date computed in the same window. In this way, we consider the market-wide movements that may coincide with downgrades and eliminate the possibility that the estimated effects are the result of changes in the US Treasury market. In particular, this excess return (ER) is computed as follows:

$$ER_i = \left(\frac{P_{t_2}^i - P_{t_1}^i}{P_{t_1}^i} - \frac{T_{t_2}^i - T_{t_1}^i}{T_{t_1}^i} \right) \frac{365}{t_2 - t_1} \quad (2)$$

where $P_{t_1}^i$ and $P_{t_2}^i$ are the last and first prices, respectively, of the rerated bond with respect to the i -event date and $T_{t_1}^i$ and $T_{t_2}^i$ are the corresponding prices of the matched Treasury security in the same event window, (t_1 , t_2). We compute the Treasury prices from the zero coupon yields of the US Treasury with the same maturity as the rerated bond. We gather the data from Bloomberg. Finally, we adjust ER by the number of days between the observed prices and convert it to an annual basis to make them comparable across events.

We analyse the effect of the split rating dynamics on the bond excess returns in a multivariate setting. We start by specifying the following baseline cross-section regression (Model 1):

$$ER_i = \beta_0 + \beta_1 CR_{i-1} + \beta_2 \Delta CR_i + \lambda W_i + \epsilon_i \quad (3)$$

where ER_i is the bond excess return around the rating event i , CR_{i-1} is the rating level of the rerated issue the day before the i -event is announced ($t = -1$) and ΔCR_i is the rating change size computed as the absolute value of the rating adjustment.¹⁶ We control for the event window size by including the number of trading days between the two prices used to estimate the excess return, that is, between t_1 and t_2 (W_i).

¹⁴ Bessembinder et al. (2009) and Ederington et al. (2015) indicate that calculating returns using clean prices, ignoring accrued interest, makes little difference in the size and power tests.

¹⁵ Each event, i , is a bond/rating event date combination. Dates are expressed relative to the event date, $t=0$.

¹⁶ $\Delta CR_i = \text{abs}(CR_i - CR_{i-1})$. We take the maximum rating variation in those cases in which multiple CRAs announce a rating event simultaneously (multi-rating events).

In order to avoid possible bias due to only analyse the effect of split rating around rating-event dates, we follow Gande and Parsley (2005) and Ferreira and Gama (2007) in creating a bond-matched sample of non-event dates for every rating event in the sample as a control group. Then, we estimate models joining both rating-event and non-event data. This approach is common in the credit rating literature (e.g., Brooks, Faff, Treepongkaruna and Wu, 2015; Abad, Alsakka and Gwilym, 2018). We create the matched sample by drawing non-event days randomly (with replacement) from the same issuer and time period¹⁷ to obtain a representative matched random sample of non-event days that replicates the main characteristics of the rating sample.¹⁸ Hence, the number of observations in the estimated models is twice the number of rating events.

To test our first hypothesis that the rating changes effects depend on the dynamics of the information opacity, we analyse the effect of the split rating dynamics (our proxy for changes in the information opacity). We consider the value of the split on the i -event date ($Split_i$) and define two dummy variables, as follows: Widen, which is equal to 1 when there is an increase in the disagreement between CRAs as a consequence of the i -event (that is, $Split_i > Split_{i-1}$) and zero otherwise; and Narrow, which is equal to 1 when the rating event diminish the disagreement ($Split_i < Split_{i-1}$). These variables signal situations in which the information opacity became higher or lower. Both dummies were added, so Model 2 is as follows:

$$ER_i = \beta_0 + \beta_1 CR_{i-1} + \beta_2 \Delta CR_i + \beta_3 Widen_i + \beta_4 Narrow_i + \lambda W_i + \epsilon_i. \quad (4)$$

To test the second hypothesis, Model 2 is extended with additive and multiplicative widen and narrow variables simultaneously to consider the split level and its dynamics jointly. Model 3 is then the following:

$$ER_i = \beta_0 + \beta_1 CR_{i-1} + \beta_2 \Delta CR_i + \beta_3 Widen_i + \beta_4 Narrow_i + \beta_5 Widen_i Split_i + \beta_6 Narrow_i Split_i + \lambda W_i + \epsilon_i \quad (5)$$

To enhance the estimation properties, we include a set of time period dummies in all models to control for the effect of time varying factors as business cycle or regulatory changes. We also include issuer dummies to control for multiple bonds from the same issuer.

4. The effects of information opacity on bond returns

We first compute average ER for both the rating-event and matched samples and test for the equality of both variables. These mean values are -0.321 , 0.082 respectively. We reject the hypothesis of equality of means and equality of medians inasmuch as the values of the corresponding t-ratio and sign test statistics equal to -8.07 and 12.83 (p-value equal to 0 in both cases). This preliminary evidence points to a significant negative effect of rating deteriorations on bond returns in line with the literature (see May, 2010; Steiner and Heinke, 2001; Jorion and Zhang, 2007).

Table 4, Panel A, presents the results of estimating Models 1, 2 and 3. The first column presents the results of Model 1, which reveal a clear pattern whereby larger sized negative rating adjustments lead to stronger decreases in bond returns. This result confirms the above preliminary evidence on the negative impact of rating deterioration on returns. Specifically, a rating deterioration of 1 point (e.g., a negative change on the watch status from any rating level) is associated with a significant drop in the excess return of -0.115 . Also, consistent with prior studies, we observe that the prior rating is relevant in the explanation of the excess return around the event. In particular, the better the prior rating level, the lower the negative impact of negative rating adjustments. The coefficient on the event-window size (the number of trading days between prices used to the compute return) is significant and presents the expected sign, which implies that the negative effect of the credit event is somewhat diluted when the returns are calculated in a broad window. However, in line with Ederington et al. (2015), this evidence indicates that the additional non-event noise over the longer periods is out-weighted by the increased sampling, allowing the negative impact of the announcement of a lower credit quality of a bond to be detected.

The second column of Panel A in Table 4 presents the results of estimating Model 2, which includes the dynamics of split rating (opacity). The investors' response to negative rating events is stronger when the announcing CRA's opinion converge with the others, which imply that information opacity decreases. A rating event that narrows the split rating leads to more negative excess returns by -0.439 . The rationale of this effect is that the reduction on the uncertainty about the creditworthiness enhance the investors reliance on the information disclosed by the rating event and induces a stronger excess return response to it. As in Model 1, the prior credit rating level and the event-window size remain significant, as their coefficients have the expected signs.

The Model 2 results support our first hypothesis, indicating that the convergence of the CRAs' opinions is important for investors along the lifetime of bonds. This result is aligned with the findings of Livingston et al. (2010) at issuance date. They prove that higher split rating levels are associated with larger yield spreads, signalling that more information opacity causes a higher cost of capital for firms. Our results complement this evidence, indicating that when a split rating converges after negative rating events, the excess return response to these events is stronger, as market participants trust the information disclosed by the rating event more.

¹⁷ We consider four periods according to the NBER US Business Cycles reference dates: The periods considered are: October 1st 2004 to December 1st 2007; December 2nd 2007 to June 1st 2009; June 2nd 2009 to June 1st 2012; June 1st 2012 to the end of the sample (see National Bureau of Economic and Research, <http://www.nber.org/cycles.html> for details).

¹⁸ The main characteristics for the matched sample (analogous to Table 2) are available on request.

Table 4
Impact of negative credit rating events on bond excess.

	Panel A			Panel B		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Intercept	−0.505 (−0.82)	−0.601 (−0.96)	−0.565 (−0.91)	0.245 (0.34)	0.078 (0.11)	0.150 (0.20)
CR _{−1}	0.041*** (5.40)	0.047*** (6.03)	0.046** (5.89)	0.051*** (4.03)	0.058*** (4.41)	0.057*** (4.29)
ΔCR	−0.115*** (−6.09)	−0.102*** (−5.04)	−0.108*** (−4.49)	−0.126*** (−6.80)	−0.113*** (−5.15)	−0.121*** (−4.91)
Widen		0.010 (0.10)	0.022 (0.13)		0.007 (0.07)	0.015 (0.10)
Narrow		−0.439*** (−3.62)	−0.739*** (−3.95)		−0.459*** (−3.94)	−0.776*** (−4.53)
Widen*Split			0.002 (0.09)			0.003 (0.15)
Narrow*Split			0.081*** (2.60)			0.087*** (2.94)
Window	0.001*** (5.60)	0.001*** (6.08)	0.001*** (6.37)	0.001*** (5.62)	0.001*** (5.96)	0.001*** (6.10)
Senior				−0.568 (−1.44)	−0.583 (−1.47)	−0.598 (−1.52)
Volume				0.000 (0.27)	0.000 (0.28)	0.000 (0.42)
SupSP				−0.223*** (−2.84)	−0.198** (−2.55)	−0.204*** (−2.61)
Covenants				−0.315 (−1.45)	−0.337 (−1.55)	−0.332 (−1.53)
Risk				0.073 (0.41)	0.095 (0.54)	0.105 (0.59)
Multi-rating				−0.100 (−1.04)	−0.103 (−1.06)	−0.089 (−0.90)
# CRA following				−0.093 (−0.69)	−0.078 (−0.57)	−0.104 (−0.78)
Maturity				−0.039 (−0.84)	−0.039 (−0.84)	−0.039 (−0.84)
Issuer & Time	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.009	0.009	0.009	0.045	0.045	0.045

The number of cases is 33,250 (16,625 credit rating deteriorations plus 16,625 non-event matched cases). All models are estimated by OLS with Huber-White robust standard errors. The t-statistics are displayed in parentheses. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. CR_{−1} is the prior rating level by the CRA announcing the event, where the subscript −1 indicates the day before the announcement. ΔCR is the size of the rating event, defined as ΔCR = abs(CR − CR_{−1}). Split is the maximum discrepancy between the ratings assigned by the CRAs. Widen (Narrow) is a dummy variables taking the value of 1 if Split > Split_{−1} (Split < Split_{−1}). Issuer is a set of dummy variables to control for the issuers. Time is a set of dummies to control for the US Business Cycle (see footnote 11¹). Window is the number of days in the event window. Senior is a dummy variable that is equal to 1 for senior issues. Volume is the volume issued of the issue affected by the event. SupSP is a dummy with a value of 1 if S&P is the CRA assigning the highest CR. Covenants is a dummy variable taking a value of 1 if the issue has some covenants. Risk is the change in the Moody's AAA Corporate Bond Index Yields. Multi-rating is a dummy variable equal to 1 for simultaneous announcements. #CRA following is the number of CRAs following the issue at the event date. Maturity is the time to maturity of the bonds at the event date.

The estimation of Model 3 is consistent with the prior models' results (see the third column, Panel A, Table 4), confirming that those negative rating events that imply the convergence across CRAs' opinions have stronger impact on excess returns than other events. Additionally, we observe milder effects when the split rating level is higher. This result supports our second hypothesis that expects weaker effects of negative rating events when there is more issuer opacity on the event date. The findings are consistent with those in Livingston and Zhou (2010), who indicate that investors are more worried about over-rating when there are high levels of split ratings. The negative impact for narrowing split rating changes diminishes until the split rating is below 10 points of divergence in the 58-point scale (one rating letter). In line with our expectations, negative rating events that imply the convergence of opinions across CRAs is the scenario in which the excess return presents a more negative response to rating deteriorations. The interpretation of this result is that this narrow split after the event signals lower uncertainty about the information that the rating adjustment discloses, making is more credible for investors. However, the impact is lower when the opacity level remains high consistently with our expectations.

To test the robustness of the previous inferences, we run again the models extended with a set of control variables. In line with Livingston et al. (2010), we control for the confounding effects of several bond characteristics such as maturity, volume, whether it is a senior bond and whether it includes covenants. To capture these characteristics we include the following set of variables for each rerated bond: Maturity is its time to maturity, Senior is a dummy variable that is equal to 1 for senior bonds, zero otherwise, Volume is the volume issued, Covenants is a dummy variable taking a value of 1 if the issue has some covenants, zero otherwise. We also control for corporate bond market conditions by including the changes in the Moody's AAA Corporate Bond Index Yields (Risk variable in the models). Finally, we include characteristics of the credit rating events,

such as the announcing CRA to allow downgrade responses to vary by rating agency. Following [Lugo et al. \(2015\)](#), we consider S&P to be the leading agency as it is the most established CRA with the highest reputational capital. We capture this by including $SupSP_i$, which is a dummy variable indicating that Standard and Poor's is assigning the highest rating to the rerated bond. Finally, we control for multi-rating events and the number of CRAs that rate the issue on the event date by including the variables $Multi-rating$, which equals 1 for simultaneous announcements, zero otherwise, and $\#CRA\ following$, that is the number of CRAs monitoring the issue at the event date.

The estimation results of Models 1, 2 and 3 extended with control variables are presented in Panel B of [Table 4](#). In general, the key results on the effect of the split rating dynamics are pervasive across models and consistent with those for their initial models' version. The coefficients of the control variables give us interesting readings. The effect of negative rating adjustments seems to be more negative for senior bonds, bonds with covenants, those affected by a simultaneous rating action, those with longer maturity and monitored for more CRAs. The effect is less negative for bonds with higher volume and when the bond market risk is higher. However, the information that these controls for individual characteristics add is scarce as their coefficients are not significant in the models. It should be noted that all models include issuer dummies, and this result seems to indicate that these dummies are capturing the effect of individual characteristics. Remarkably, the negative rating events affecting bonds for whom S&P is the CRA assigning the best rating cause a significantly stronger negative response of excess returns, pointing to a leading position of S&P in the US market. This finding is consistent with those presented by [Livingston and Zhou \(2010\)](#), who show that bonds with superior S&P ratings present less penalized yields due to split ratings.

5. Further analysis

5.1. Investment-Grade versus High-Yield issues

The literature suggests that investors are more sensitive to negative rating adjustments of speculative issues than of IG ones, showing a stronger reaction in the former case ([Jorion and Zhang, 2007](#); [Cantor and Packer, 1995](#)). Some institutional investors face regulatory restrictions that mainly focus on a broad ratings category, i.e., the IG segment, rather than individual letter grades or notches within a category (e.g., [Cantor, Gwilym and Thomas, 2007](#); [Baghai, Becker and Pitschner, 2019](#)). Regulation commonly bans the investment in speculative bonds so, rating events could reflect trading for regulatory reasons rather than reflecting the information content of rating adjustments. Furthermore, speculative bonds are more sensitive to firm-specific information ([Kwan, 1996](#); [Han and Zhou, 2013](#)).

So, we deepen the investigation into the rating level aspect of opacity by studying the effects of the split rating dynamics on IG and HY issues separately. The IG group comprises bonds with at least a BBB- rating after the negative rating event. The HY group comprises issues that have BB + or lower ratings before the event. We restrict the group to those cases in which the rerated bond remains in the same segment (IG or HY) after the rating event, i.e., we exclude the "fallen angels" from the sample. This allows to avoid the contamination induced by considering bonds that crosses the IG-HY boundary, more strongly affected by regulatory-induced investors' response.

[Table 5](#) shows the distribution of rating events and split ratings separately for both groups. The IG group presents 10,698 events, more than twice the number of HY events (4,798). The average rating deterioration for IG is 3.8 points on the scale, slightly below the average of the HY group (4.3 points). There is also a difference in the split rating level, as the average split rating level of the IG group is lower than that of the HY group on the event day. It is important to highlight that the percentage of HY rerated bonds showing a split rating is much higher (97%) than in the case of the IG group, which is 75%. [Livingston, Naranjo and Zhou \(2007\)](#) find that IG bonds are less likely to have a split rating than HY bonds. [Livingston et al. \(2008\)](#) show that highly rated bonds (AAA and AA) are less likely to have rating changes than BBB rated bonds. The average change in the split rating (i.e., in information opacity) is similar in both samples (0.81 and 0.85 points, respectively). In both groups, the percentage of rating adjustments related to opacity increases (widen split rating) is higher than the percentage of those leading to lower information opacity (narrow split rating). The narrows average size is higher in the HY group whereas the average size of the widens is higher in the IG group. However, the percentage of widen events is higher in the HY group than in the IG group, indicating a higher percentage of events related to increases in the opacity level for the lower quality bonds.

We test our third hypothesis by running Models 1, 2 and 3 separately for both groups, excluding the CR_{-1} variable. [Table 6](#) presents the results for HY and IG bonds. Overall, our findings document the relevance of the credit risk level, indicating that the HY segment is more sensitive to opacity than the IG segment.

The key results observed for the whole sample (see [Table 4](#)) are maintained for the HY group (see [Table 6](#)). Rating deteriorations lead to significant negative excess returns for low rated bonds and this response depends on the size of the rating change, showing a stronger impact for larger rating falls. Downgrades leading to narrower split rating produce more negative excess returns and their impact is stronger than for the whole sample. This impact is reduced for higher split rating levels after the event date, as in the general case, resulting in a positive impact for those with a split rating of more than 7 points. Additionally, rating deteriorations increasing the split rating (that is, widening rating-events) lead to lower negative return.

Remarkably, the key results observed for the whole sample are not maintained for the IG group ([Table 6](#)). Negative rating events on investment grade issues induce negative excess returns linked to the size of the rating changes, but the dynamic of the information opacity of the issuers is not relevant. Investors seem to consider the uncertainty about the creditworthiness

Table 5
CR and split rating distribution by rating level.

	Investment Grade (N = 10,698)					High Yield (N = 4,798)				
	#	(%)	Mean	Max	Min	#	(%)	Mean	Max	Min
ΔCR	10,698	64%	3.78	16	1	4,798	29%	4.34	17	1
Splitt	8,022	75%	5.44	25	0	4,672	97%	7.71	44	0
Splitt ₋₁	8,061	75%	4.68	30	0	4,678	97%	6.99	44	0
ΔSplitt	7,272	68%	0.81	15	-10	3,949	82%	0.85	15	-24
ΔSplitt = 0	3,426	32%				849	18%			
ΔSplitt < 0 (Narrow)	3,001	28%	-2.78	-1	-10	1,401	29%	-3.10	-1	-24
ΔSplitt > 0 (Widen)	4,271	40%	3.33	15	1	2,548	53%	3.02	15	1

See the note for Table 3. *Investment Grade* includes cases in which the rerated bond remains at the IG level after the negative rating adjustment (that is, that became rated above BBB-). *High Yields* includes cases in which the rerated bond was HY before the negative rating adjustment (that is, between BB + and D). The percentages are calculated based on the total cases in the corresponding subsample.

of the issuer reflected by the split rating not relevant when there is low credit risk, i.e., for bonds in the investment segment of the market.

Interestingly, the impact of credit events in which Standard and Poor's is the CRA assigning the highest grade becomes significant only for the IG sample, indicating that the market relies more on S&P in the investment segment. As Moody's provides more conservative ratings than S&P, the downgrades by Standard and Poor's only lead to stronger effects in the low credit risk segment.

5.2. Alternative rating disagreement measures

In the previous sections, we measured the CRAs' disagreement with the standard split rating measure in the literature given in expression (1). However, most issues in our sample are rated by three agencies (63.6%) and this measure ignores this information. The split rating measure only considers the two most extreme ratings, while ignoring the information of the ratings that are in the middle. This information could be relevant to determine the importance of the disagreement among CRAs. In one hand, the standard measure does not allow relevant cases to be distinguished, such as those when there is a divergence of opinion between all the CRAs (three different credit ratings for the same bond) from the case when two CRAs share the same rating opinion and the third disagrees. However, the first case implies a higher level of uncertainty regarding the creditworthiness of the issue than the second case. In the other hand, the regulatory use of credit ratings is often non-linear and may depend in complex ways on the interaction of the agencies. For instance, some investors look at median ratings, some other at the lowest rating. The split measure ignores these effects.

To overcome these shortcomings, we propose two alternative measures of disagreement that consider not only the extreme ratings but the entire rating distribution. The first is the Euclidean distance between the CR assigned by the CRAs (ED hereafter), which is computed as follows:

$$ED_i = \sqrt{(CR_{i,S\&P} - CR_{i,Moodys})^2 + (CR_{i,S\&P} - CR_{i,Fitch})^2 + (CR_{i,Fitch} - CR_{i,Moodys})^2} \quad (6)$$

where $CR_{i,S\&P}$, $CR_{i,Moodys}$ and $CR_{i,Fitch}$ are the credit rating (into a 58-point numerical scale) of the issue by S&P, Moody's and Fitch, respectively, on the i -event date.

The second measure is the Euclidean distance to the worst rating (EDW, hereafter), which we define as follows:

$$EDW_i = \sqrt{(CR_{i,S\&P} - WCR_i)^2 + (CR_{i,Moodys} - WCR_i)^2 + (CR_{i,Fitch} - WCR_i)^2} \quad (7)$$

where $WCR_i = \min(CR_{i,S\&P}, CR_{i,Moodys}, CR_{i,Fitch})$. ED considers the three ratings equally, whereas EDW considers the worst rating as the reference point.

We repeat all the analysis by replacing the split variable by these new measures each time. Table 7 presents the results for ED and EDW separately. Noticeably, the key results obtained for the whole sample (Table 4) are mostly the same as the results with the ED measure, as this also happens when considering the distance to the worst measure, EDW (Table 7).

The main results with our two new proxies of CRAs' disagreement points to the robustness of our main conclusions along the paper. They allow to discard that the observed ER responses to split rating dynamics are related to the presence of non-linear regulatory uses of rating. This reinforces the evidence we find on the relevance the firms' information opacity dynamics during the entire period that bonds are in the market and not only at issuance (see Livingston and Zhou, 2010).

Table 6
Impact of credit rating deteriorations on bond excess returns: High Yield (HY) and Investment Grade (IG).

	Panel A						Panel B					
	Model 1		Model 2		Model 3		Model 1		Model 2		Model 3	
	HY	IG	HY	IG	HY	IG	HY	IG	HY	IG	HY	IG
Intercept	-0.805* (-1.90)	-0.021 (-0.04)	-0.876** (-2.01)	-0.038 (-0.07)	-0.570 (-1.24)	-0.034 (-0.06)	4.202* (1.71)	-0.075 (-0.10)	3.793 (1.58)	-0.058 (-0.07)	4.599** (1.96)	-0.061 (-0.08)
ΔCR	-0.200*** (-6.17)	-0.058* (-2.02)	-0.139*** (-2.62)	-0.069** (-2.56)	-0.190*** (-3.17)	-0.067** (-2.32)	-0.227*** (-6.17)	-0.086*** (-4.30)	-0.151*** (-2.75)	-0.098*** (-3.39)	-0.209*** (-3.34)	-0.090*** (-3.55)
Widen			-0.175 (-0.89)	0.136 (1.24)	0.766** (2.29)	0.196 (0.89)			-0.161 (-0.82)	0.139 (1.14)	0.624* (1.88)	0.352 (1.22)
Narrow			-0.786** (-2.28)	-0.022 (-0.19)	-2.758*** (-4.08)	-0.145 (-0.80)			-1.020*** (-2.99)	-0.023 (-0.19)	-2.887*** (-4.18)	-0.133 (-0.71)
Widen*Split					-0.093** (-2.43)	-0.011 (-0.38)					-0.074** (-2.0)	-0.038 (-1.02)
Narrow*Split					0.447*** (3.48)	0.036 (1.10)					0.449*** (3.30)	0.026 (0.79)
Window	0.002*** (3.04)	0.001*** (5.08)	0.002*** (2.98)	0.001*** (5.00)	0.002*** (2.97)	0.001*** (5.03)	0.001*** (2.38)	0.001*** (4.60)	0.001*** (2.31)	0.001*** (4.50)	0.001*** (2.33)	0.001*** (4.78)
Senior							-1.360 (-0.58)	-0.583 (-1.42)	-1.147 (-0.50)	-0.589 (-1.42)	-1.262 (-0.58)	-0.595 (-1.44)
Volume							0.000 (0.86)	0.000 (0.26)	0.000 (0.92)	0.000 (0.29)	0.000 (1.12)	0.000 (0.29)
SupSP							0.255 (1.43)	-0.227** (-2.14)	0.450*** (2.67)	-0.228** (-2.16)	0.164 (0.93)	-0.231** (-2.24)
Covenants							-0.141 (-0.50)	-0.440 (-1.44)	-0.174 (-0.62)	-0.436 (-1.45)	-0.216 (-0.76)	-0.441 (-1.47)
Risk							-1.387*** (-5.80)	0.349 (1.39)	-1.460*** (-6.22)	0.348 (1.41)	-1.329*** (-5.91)	0.344 (1.39)
Multi-rating							0.400* (1.66)	-0.368*** (-2.68)	0.307 (1.24)	-0.373*** (-2.64)	0.337 (1.35)	-0.383** (-2.54)
# CRA following							-1.423*** (-3.72)	-0.002 (-0.01)	-1.414*** (-3.69)	-0.016 (-0.10)	-1.685*** (-4.12)	-0.004 (-0.03)
Maturity							0.010*** (23.43)	-0.059 (-0.91)	0.010*** (23.52)	-0.059 (-0.91)	0.010*** (23.47)	-0.059 (-0.91)
Issuer & Time	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.056	0.007	0.058	0.007	0.062	0.007	0.067	0.079	0.069	0.079	0.073	0.079

See the note for Table 4. The number of cases is 9,596 (4,798 credit rating deteriorations plus 4,798 non-event matched cases) on the High Yield (HY) sample, that is, from BB + watch + to D and 21,396 (10,698 credit rating deteriorations plus 10,698 non-event matched cases) on the Investment Grade (IG) sample, that is, from AAA to BBB + watch --.

Table 7
Impact of rating deteriorations on bond excess returns. Euclidean distance and Distance to the worst.

	Panel A				Panel B			
	Model 2		Model 3		Model 2		Model 3	
	ED	EDW	ED	EDW	ED	EDW	ED	EDW
Intercept	−0.569 (−0.93)	−0.516 (−0.84)	−0.523 (−0.85)	−0.480 (−0.78)	0.129 (0.18)	0.234 (0.32)	0.228 (0.31)	0.311 (0.42)
CR _{−1}	0.047*** (6.02)	0.043*** (5.60)	0.045*** (5.77)	0.044*** (5.67)	0.058*** (4.40)	0.053*** (4.19)	0.056*** (4.25)	0.055*** (4.28)
ΔCR	−0.117*** (−5.44)	−0.121*** (−5.54)	−0.122*** (−4.82)	−0.137*** (−5.26)	−0.127*** (−5.40)	−0.132*** (−5.43)	−0.135*** (−5.04)	−0.151*** (−5.36)
Widen _{ED}	0.157 (1.49)		0.258* (1.74)		0.150 (1.48)		0.247* (1.90)	
Narrow _{ED}	−0.266** (−2.19)		−0.685*** (−3.94)		−0.293** (−2.51)		−0.727*** (−4.58)	
Widen _{EDW}		0.124 (1.12)		−0.009 (−0.06)		0.125 (1.15)		−0.013 (−0.09)
Narrow _{EDW}		−0.053 (−0.47)		−0.332** (−2.16)		−0.064 (−0.58)		−0.348** (−2.47)
Widen _{ED} *ED			−0.008 (−0.58)				−0.006 (−0.46)	
Narrow _{ED} *ED			0.078*** (3.76)				0.083*** (4.17)	
Widen _{EDW} *EDW				0.024 (1.29)				0.027 (1.48)
Narrow _{EDW} *EDW				0.061*** (3.30)				0.065*** (3.68)
Window	0.001*** (5.88)	0.001*** (5.70)	0.001*** (6.31)	0.001*** (6.00)	0.001*** (5.81)	0.001*** (5.53)	0.001*** (6.13)	0.001*** (5.70)
Senior					−0.583 (−1.47)	−0.575 (−1.46)	−0.600 (−1.52)	−0.593 (−1.50)
Volume					0.000 (0.24)	0.000 (0.26)	0.000 (0.42)	0.000 (0.41)
SupSP					−0.208*** (−2.67)	−0.221*** (−2.83)	−0.215*** (−2.72)	−0.200** (−2.51)
Covenants					−0.332 (−1.53)	−0.316 (−1.47)	−0.318 (−1.47)	−0.300 (−1.40)
Risk					0.094 (0.53)	0.075 (0.43)	0.105 (0.59)	0.088 (0.50)
Multi-rating					−0.116 (−1.19)	−0.110 (−1.13)	−0.112 (−1.14)	−0.111 (−1.09)
# CRA following					−0.087 (−0.63)	−0.095 (−0.67)	−0.130 (−0.93)	−0.154 (−1.09)
Maturity					−0.039 (−0.84)	−0.039 (−0.84)	−0.039 (−0.84)	−0.039 (−0.84)
Issuer & Time	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.009	0.009	0.009	0.009	0.045	0.045	0.045	0.045

See the note for Table 4. $ED = \sqrt{(CR_{S\&P} - CR_{Moody's})^2 + (CR_{S\&P} - CR_{Fitch})^2 + (CR_{Fitch} - CR_{Moody's})^2}$ is the Euclidean distance of the ratings between all CRA following the issue. $Widen_{ED}$ ($Narrow_{ED}$) is a dummy variable taking the value of 1 if $ED > ED_{-1}$ ($ED < ED_{-1}$). $EDW = \sqrt{(CR_{S\&P} - WCR)^2 + (CR_{S\&P} - WCR)^2 + (CR_{Fitch} - WCR)^2}$, where $WCR = \min(CR_{S\&P}, CR_{Moody's}, CR_{Fitch})$, is the Euclidean distance of each CRA's rating to the worst rating (WCR). $Widen_{EDW}$ ($Narrow_{EDW}$) is a dummy variable taking the value of 1 if $EDW > EDW_{-1}$ ($EDW < EDW_{-1}$).

6. Conclusions

Based on a sample of 16,625 announcements of rating deterioration of US corporate bonds rated by Standard and Poor's, Moody's and Fitch ratings between 2004 and 2014, we examine the dynamic evolution of the CRAs' disagreement on the creditworthiness (split rating) as a measure of changes in the information opacity of issuers. According to previous literature, we confirm that the largest rating deteriorations lead to more negative bond return responses, but this effect depends strongly on the information opacity of the issue. Therefore, we contribute to the extant literature by considering the effect of the dynamics of a split rating on corporate bond returns, an issue that has not been previously addressed in the corporate credit ratings analysis.

Our investigation on the role of information opacity on the bond returns response to negative credit rating events reveals that both the level and the dynamics of the information opacity (i.e., increases and decreases in the split rating) are important to explain this response. When the rating deterioration reflects a reduction in the information opacity, the excess return

response is more intense than in other cases. This response is less intense for bonds that show a higher opacity level at the event date. The split rating acts as a signal for investors who perceive higher uncertainty regarding the issue credit risk for high disagreement. This situation lessens the credibility of the information disclosed by rating changes. These results highlight the role of the issuer's information opacity on the bond price construction.

An interesting result is that the effect of information opacity is asymmetric depending on the credit quality of the rerated bonds. The evidence indicates that this phenomenon is relevant for High Yield debt, but it is not for bonds in the investment segment. This suggests that investors are less concerned about the uncertainty regarding the creditworthiness of higher rated issues.

Finally, we show that the key results are robust when we consider two alternative measures to estimate the rating divergence of opinion. We define two measures that consider all the credit scores assigned to a bond instead of only the two extreme ratings needed to compute the split rating measure. These measures could be helpful for further analysis on the CRAs' divergence of opinion, as they capture any change in the rating distribution.

Our results complete the analysis of the effects of disagreement among CRAs in the corporate bond market considering the whole lifetime of the bond. These results are useful to investors, issuers and capital market regulators. The results highlight that issuers should consider the effects of the opacity dynamics on the cost of their debt, as it affects the returns of their bonds during their lifetime. The dynamics of CRAs' disagreement could not only penalize the debt in secondary markets, but this would also be relevant for the primary market. Bond investors should consider the opacity of firms to manage the credit risk of their portfolios. Finally, regulators could use the dynamics of split ratings to monitor the information opacity of firms and could design regulations that reinforce transparency as a tool to enhance the assessment of credit risk in the bond market.

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